Péter Lódri

Systems engineer & independent researcher — Rust, verifiable computation, distributed infrastructure

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Summary

Rust-first systems engineer focused on cryptographic provenance and verifiable computation: hash-verified ledgers, signed receipts, challenge-response clock proofs, and replay/verification gates. The same discipline ZK proving and L2 correctness depend on — prove it, don't assert it. I run a self-hosted NixOS / nix-darwin fleet (30+ services) and ship open source across ~70 repos.

Skills

- Languages: Rust (primary), Nix, Python, TypeScript, Swift, Go
- Systems: NixOS / nix-darwin (3 hosts + Mac, 30+ services), Docker/OCI, Tailscale, sops-nix secrets, self-hosted CI runners
- Cryptography & verification: Ed25519, BLAKE3, Merkle trees, signed receipts, hash-verified ledgers, challenge-response clock proofs, replay/verification gates
- Distributed: mesh relay (WebSocket + QUIC), event replay, monotonic ledgers, capability-gated brokers
- Quantization: BitNet b1.58 ternary ({-1, 0, +1}) inference, MLX Metal GPU kernels, perplexity benchmarking

Selected work

- mem8 / MEM|16-10 — verifiable memory: identity as continuity evidence, hash-verified ledger, no-refunds execution budget. Pure Rust, #![no_std], zero-alloc.
- MLX-QUANT — Apple Silicon MLX low-bit ternary (BitNet b1.58) quantization, Metal kernels, perplexity benchmarking.
- 8b-is-engine — mesh relay (WS/QUIC), node framing/identity/replay boundaries, world dashboard.
- nix-base — NixOS / nix-darwin fleet managing 3 Hetzner hosts + a Mac, ~30 services, secrets via sops-nix.

Experience

Independent researcher & systems engineer — self-directed, open source. (Detailed employment history available on request.)

Interests

Based rollups, ZK proving, L2 infrastructure, sovereign systems, verifiable computation, low-bit inference.